

KoinWorks: Overcoming cultural boundaries to improve financial access and education for MSMEs

Tell us your challenge. We're here to help.

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ABOUT KOINWORKS



About KoinWorks

KoinWorks' engine quickly scores MSMEs to qualify for loans and offers a variety of financial products. The speed, flexibility, and cost-efficiency of microservices.

KoinWorks is the first credit-led Micro, Small, and Medium Enterprises (MSMEs) neobank in Indonesia that provides business development solutions. It has over 2 million users, including freelancers and startups that are just beginning their business journeys. Founded in 2016, KoinWorks originally started as a digital lending company, and continues to support the growth of the economy by helping Indonesians achieve their financial goals with financial education.

Google Cloud results

- Grew from 20 staff to 900 within 6 years to serve 2 million users
- Score SME creditworthiness through its internal scoring model within minutes
- Scaled a new product's capacity to handle traffic as it rose from 4 to 8 terabytes per month

What happens to Indonesian Medium Enterprises (MSMEs) with a loan but getting one is seen as rare where MSMEs financing is further collateralized lending, online business to [KoinWorks](https://koinworks.co) (https://koinworks.co) Indonesian financial superapp. 1 digital financial services company address this challenge by simplifying access and improving financial

KoinWorks
scaling
superapp
2 million

Industries: Financial Services & Insurance

Location: Indonesia

(https://cloud.google.com/kubernetes-engine/).



About Devoteam

Devoteam is a technology consultancy with core competencies in digital strategy, platform technologies, and cybersecurity. It helps organizations deepen their understanding on how to best leverage key tech solutions to scale their business.

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Google Cloud (<https://cloud.google.com/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Tell us your challenge. We're here to help.
Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

About Devoteam

It currently serves over two million MSMEs in Indonesia as a one-stop shop, offering lending, investment, neobanking and other products in its superapp. With a client list that includes Tokopedia and Shopee in its client list, KoinWorks has come a long way in transforming the MSME financial landscape in Indonesia. However, this transformation is not just a result of an innovative approach to lending, but also a realization through data.

Google Workspace

(<https://workspace.google.com/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

"In our initial year, KoinWorks worked exclusively with MSMEs because they have a lot of data that we can use to underwrite their loans. This data includes transaction data, total warehouse inventory, and more, that mimics know-

your-customer
and know-your-
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that banks
usually do."

—Jonathan Bryan, Chief
Platform Officer,
KoinWorks

Opening new paths for MSMEs growth

"How can we underwrite MSMEs loans? That's the biggest question," says Jonathan Bryan, Chief Platform Officer at KoinWorks.

"In our initial year, KoinWorks worked exclusively with ecommerce MSMEs because they have a lot of data that we can use to underwrite their loans. This data includes transaction data, total warehouse inventory, and more, that mimics know-your-customer and know-your-business checks that banks usually do," he adds.

These data points are fed into KoinWorks' scoring model that covers the five Cs, character, capability, condition, economic capital, and collateral. The model also factors creditworthiness through data pulled from the Directorate of Population and Civil Registration (Dukcapil), Indonesian credit bureaus,

and open sources like [Google Business Profile](https://www.google.com/business/) (<https://www.google.com/business/>).

The ability to productionize the scoring model, however, was a significant challenge for the startup. It needed the right cloud platform that could help it scale its product offerings and ensure fast scoring to make the loan application experience as seamless as possible for MSMEs.



Finding the right partner for national ambitions

When KoinWorks wanted to grow its cloud-based tech stack, there were several key considerations for the team, encompassing regulatory requirements and the best use for their workflows.

"The first consideration was that we needed our data to be stored in Indonesia as that's required by the financial authorities. Second, we needed a cloud counterpart who really understood the importance of security because we store a lot of personal and transactional data. The third was that we needed to be able to extract this data very quickly so we can do credit scoring within minutes," Bryan explains.

The KoinWorks team discovered that Google Cloud could meet these three key requirements. Although the team had previously built their superapp on top of a legacy cloud-as-a-service platform, they were growing rapidly and needed to overcome computing chokepoints and scale more cost-effectively to serve hundreds of thousands of MSMEs.

The possibilities offered by microservices on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) appeared as the right solution to help them accomplish their growth plans moving forward. As a result, Google Cloud was determined as the best fit for its engineers as it was "the home of Kubernetes," according to KoinWorks' Head of Infrastructure, Mirasz Albari.

"With GKE, we can enable cluster auto-scaling. So if we determine that one or several products or features are underperforming, we can easily review and reconfigure its utilization to suit our requirements. It's also easy to stress test our system to find the right utilization amount for features and products," explains Albari.



Scaling a superapp at minimal cost

Without auto-scaling, the alternative for KoinWorks' engineers would be to manually redeploy over 80 microservices in KoinWorks, which would render the app inactive.

"If users are unable to access our superapp and complaints start coming to our customer service team, that's normal. But the worst outcome would be if complaints are published on social media, and go viral. So even though service disruption may last a short while, it can be blown out of proportion when it becomes highlighted on social media," Albari warns.

"Integration with Google Cloud has had a very big impact on our infrastructure. Since introducing KoinWorks NEO in April 2022, overall traffic to KoinWorks has increased from three to four terabytes per month, to seven to eight terabytes," Albari reveals.

The success of KoinWorks NEO , the first MSME neobank in Indonesia, and the positive response from end-users have been directly attributed by KoinWorks' infrastructure team to GKE's ability to auto-scale node pools to meet customer demand during a new product launch.

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features and
products."

*—Mirasz Albari, Head of
Infrastructure, KoinWorks*

Google Cloud's ability to segment billing on a per-project basis has also proved beneficial for the company. As KoinWorks grew, it split up into different entities and could map billing and budgeting to each entity. As a result, KoinWorks could isolate projects to different entities and determine product operating costs with ease.

"The bottom line for any startup is that costs need to be optimized. We can also keep track of urgent expenditures, determine why certain compute costs are high, and split our billing when provisioning our superapp," he adds.

As a result, the team can determine how costs tally with key metrics like user engagement and transaction volumes for each product like KoinBisnis, KoinP2P, KoinRobo, and more.

Coincidentally, the alignment between GKE's autoscaling capabilities and KoinWorks' decision to split its engineers into product groups has enabled cost controls at a more granular level. But deeper

integration with Google Cloud has also opened up new opportunities to grow.

Leveraging Indonesia's affinity for Google services

The ease through which KoinWorks is able to source local talent familiar with Google Cloud has helped it accelerate app delivery to its MSMEs customer base. Thanks to Google Cloud's strong community in Indonesia, the onboarding process for fresh engineering talent is simplified because they are familiar with the platform and technologies like GKE and Cloud SQL (<https://cloud.google.com/sql>).

For example, KoinWorks' engineers can create their own CI/CD to deploy in Google Cloud and automate app development. This advantage is key to delivering a superapp and rolling out a comprehensive business ecosystem for MSMEs where they can improve their financial literacy. Rich new features that have been added to the business ecosystem include learning modules on accounting basics, financial management, digital marketing, and more. MSMEs can also pay their bills on the superapp. These services are a part of the strategy by the KoinWorks team to retain customers on the app.

"We are building an ecosystem where a user can do everything in one place, from starting their business to growing it. Currently, we're working on a feature where they will eventually be able to sell their business to an investor through our app," states Bryan. "We also utilize Google solutions as much as

possible because we know that MSMEs are deeply integrated with Google Workspace products like Gmail (<https://workspace.google.com/products/gmail/>) and Calendar (<https://workspace.google.com/products/calendar/>) by default."

"KoinWorks can become more than just a peer-to-peer lender with all these products in our ecosystem. We can help MSMEs with Google solutions in parallel with Google Indonesia's efforts to help local businesses digitize."

—Jonathan Bryan, Chief
Platform Officer,
KoinWorks

Empowering nanopreneurs with financial access

KoinWorks has its sights firmly set on expanding service delivery to underserved demographics in Indonesia, including a larger segment of Indonesian MSMEs like the ubiquitous *warungs*, and the content creator and freelancer markets.

"I call them the blue ocean for lending and it is also one of the most appealing segments in Indonesia," shares Bryan.

As they blaze the path in financial access, the KoinWorks team is excited to make fuller use of its own data to score MSMEs instead of relying on partner data.

"KoinWorks can become more than just a peer-to-peer lender with all these products in our ecosystem. We can help MSMEs with Google solutions in parallel with Google Indonesia's efforts to help local businesses digitize," he concludes.